THE CHINESE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS
MATH1025/1028 Essential Mathematical Methods 2025-26 First Term

Solution to Homework 10

Complusory Questions

1. Express each of the following curves by an equation in x and y, then sketch it and indicate the starting point, endpoint and direction of increasing t.

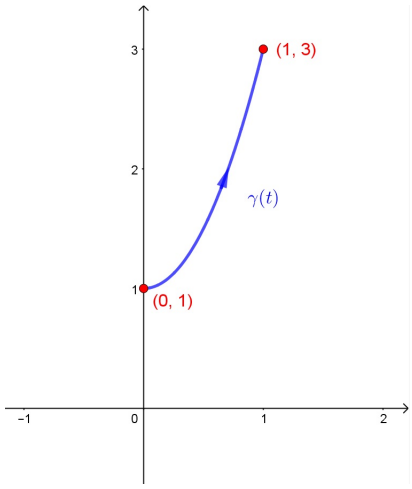
   (a) $\gamma(t) = (x(t), y(t)) = (\sqrt{t}, 2t+1)$ for $0 \le t \le 1$;

   (b) $\gamma(t) = (x(t), y(t)) = (t, \frac{4}{t})$ for $1 \le t \le 4$.

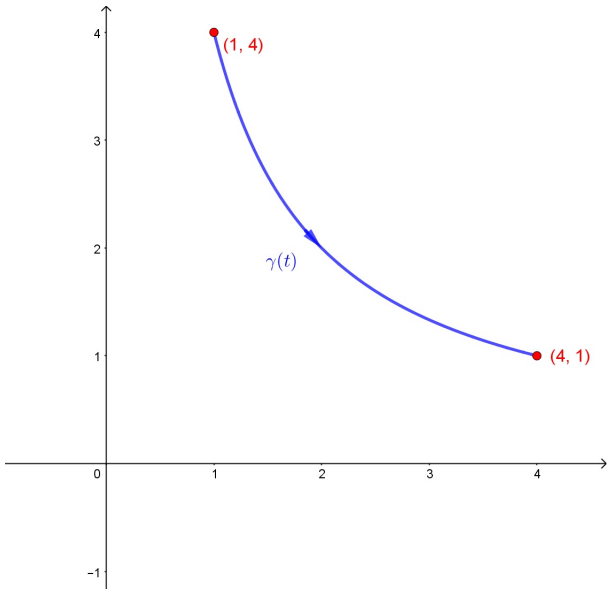
   (c) $\gamma(t) = (x(t), y(t)) = (-\cos t, \sin t)$ for $0 \le t \le \pi$.

   Ans:

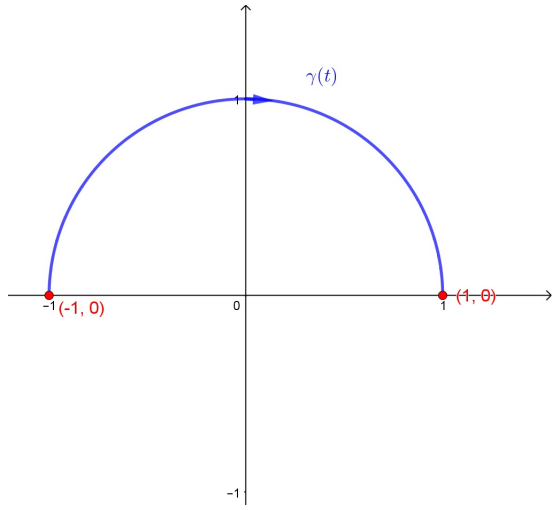
   (a) $x = \sqrt{t}$ implies that $x^2 = t$, and so $y = 2x^2 + 1$.



   (b) $x = t$ implies $y = \frac{4}{t} = \frac{4}{x}$, and so $y = \frac{4}{x}$.



(c) We take squares of both given equations and add them up, we have $x^2 + y^2 = (-\cos t)^2 + (\sin t)^2 = \cos^2 t + \sin^2 t = 1$, and so $x^2 + y^2 = 1$.



2. Let $\mathcal{P} : y^2 = 4ax$ be a parabola, where $a > 0$.

(a) Show that the equation of normal to the parabola $\mathcal{P}$ at the point $(at^2, 2at)$ is

$$y + tx = 2at + at^3.$$

(b) Given that a circle intersects the parabola $\mathcal{P}$ at four distinct points $(at_i^2, 2at_i)$, where $i = 1, 2, 3, 4$. Show that $t_1 + t_2 + t_3 + t_4 = 0$.

Ans:

(a) $\dfrac{dy}{dx} = \dfrac{\frac{dy}{dt}}{\frac{dx}{dt}} = \dfrac{2a}{2at} = \dfrac{1}{t}$. Therefore the slope of the required normal is $-t$. The equation of the required normal is

$$\begin{aligned} \frac{y - 2at}{x - at^2} &= -t \\ y + tx &= 2at + at^3 \end{aligned}$$

(b) Suppose that $x^2 + y^2 + Dx + Ey + F = 0$ is a circle that intersects the parabola $\mathcal{P}$ at four distinct points $(at_i^2, 2at_i)$, where $i = 1, 2, 3, 4$. Then,

$$\begin{aligned} (at_i^2)^2 + (2at_i)^2 + D(at_i^2) + E(2at_i) + F &= 0 \\ a^2t_i^4 + (4a^2 + Da)t_i^2 + (2Ea)t_i + F &= 0 \end{aligned}$$

Therefore, t_1, t_2, t_3, t_4 are the roots of the equation

$$a^2t^4 + (4a^2 + Da)t^2 + (2Ea)t + F = 0.$$

Sum of the roots of the above equation is $t_1 + t_2 + t_3 + t_4 = -\dfrac{0}{a^2} = 0$.

3. Let $\mathcal{E} : \dfrac{x^2}{25} + \dfrac{y^2}{16} = 1$ be an ellipse.

(a) Show that the equation of tangent to the ellipse $\mathcal{E}$ at the point $(5\cos\theta, 4\sin\theta)$ is

$$(4\cos\theta)x + (5\sin\theta)y = 20.$$

(b) For $0 < \theta < \dfrac{\pi}{2}$, the tangent to the ellipse at the point P intersects the x-axis and the y-axis at R and S, respectively. Find the least possible area of the triangle ORS.

Ans:

(a) $\dfrac{dy}{dx} = \dfrac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \dfrac{4\cos\theta}{-5\sin\theta}$. The equation of the required tangent is

$$\begin{aligned} \frac{y - 4\sin\theta}{x - 5\cos\theta} &= \frac{4\cos\theta}{-5\sin\theta} \\ (4\cos\theta)x + (5\sin\theta)y &= 20 \end{aligned}$$

(b) Put $y = 0$ and $x = 0$ into the equation obtained in (a), then we have $R = (\dfrac{5}{\cos\theta}, 0)$ and $S = (0, \dfrac{4}{\sin\theta})$, and so the area of triangle ORS is $\dfrac{10}{\cos\theta\sin\theta} = \dfrac{20}{\sin 2\theta} \geq 20$. The least possible area of the triangle ORS is 20, which is attained when $\theta = \dfrac{\pi}{4}$.

4. The straight line $y = mx + c$ intersects the ellipse $x^2 + 4y^2 = 16$ at two points P and Q.

(a) Suppose that M is the mid-point of P and Q. Show that $M = \left(-\dfrac{4mc}{4m^2+1}, \dfrac{c}{4m^2+1}\right)$.

(b) If P and Q move along the ellipse in a way that the chord PQ always passes through the point $(2, 0)$. Show that M lies on the ellipse $x^2 + 4y^2 = 2x$.

Ans:

(a) Let $P = (x_1, y_1)$ and $Q = (x_2, y_2)$. Then, $M = \left(\dfrac{x_1 + x_2}{2}, \dfrac{y_1 + y_2}{2}\right)$.

Put $y = mx + c$ into the equation of the ellipse $x^2 + 4y^2 = 16$. Then,

$$\begin{aligned} x^2 + 4(mx + c)^2 &= 16 \\ (4m^2 + 1)x^2 + 8mcx + (4c^2 - 16) &= 0 \end{aligned}$$

Note that x_1 and x_2 are the roots of the above equation. Therefore, $x_1 + x_2 = \dfrac{-8mc}{4m^2+1}$, and $y_1 + y_2 = m(x_1 + x_2) + 2c = \dfrac{2c}{4m^2+1}$. Therefore, $M = \left(-\dfrac{4mc}{4m^2+1}, \dfrac{c}{4m^2+1}\right)$.

(b) Since the chord PQ always passes through the point $(2, 0)$, $2m + c = 0$, i.e. $c = -2m$. Then, we have $M = \left(\dfrac{8m^2}{4m^2+1}, -\dfrac{2m}{4m^2+1}\right)$. Let $x = \dfrac{8m^2}{4m^2+1}$ and $y = -\dfrac{2m}{4m^2+1}$. Then,

$$\begin{aligned} \frac{x}{y} &= \frac{8m^2}{-2m} \\ m &= -\frac{x}{4y} \end{aligned}$$

and

$$\begin{aligned} y &= -\frac{2m}{4m^2+1} \\ y &= -\frac{2(-\frac{x}{4y})}{4(-\frac{x}{4y})^2 + 1} \\ y &= \frac{2xy}{x^2 + 4y^2} \\ x^2 + 4y^2 &= 2xy \end{aligned}$$

5. A straight line of slope m passing through $(1, 1)$ is tangent to the hyperbola $\mathcal{H} : xy = 2c^2$.

(a) Show that $m^2 + (8c^2 - 2)m + 1 = 0$.

(b) Hence, show that when $2c^2 > 1$, there are two tangents to $\mathcal{H}$ which pass through the point $(1, 1)$ and that the product of their slopes is independent from c.

Ans:

(a) $y-1=m(x-1)$ is a straight line of slope m passing through $(1,1)$. Put $y=mx-m+1$ into the equation of the hyperbola $xy=2c^2$, and we get

$$\begin{aligned} x(mx-m+1) &= 2c^2 \\ mx^2+(1-m)x-2c^2 &= 0 \end{aligned}$$

If the straight line is tangent to the hyperbola, then the above quadratic equation only has one real root. Therefore,

$$\begin{aligned} (1-m)^2-4m(2c^2) &= 0 \\ m^2+(8c^2-2)m+1 &= 0 \end{aligned}$$

(b) Consider the quadratic equation in m obtained in (a), since $\Delta=(8c^2-2)^2-4(1)(1)=64c^4-32c^2=32c^2(2c^2-1)>0$, it has two distinct roots. It implies there are two tangents to $\mathcal{H}$ which pass through the point $(1,1)$. Furthermore, the product of the slopes of the two tangents is the product of roots, which is 1, independent from c.

6. Let $\mathcal{E}:\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=1$ be an ellipse.

(a) (i) Show that the equation of the chord AB of $\mathcal{E}$ whose mid-point is $P(\alpha,\beta)$ is

$$\frac{\alpha x}{a^2}+\frac{\beta y}{b^2}=\frac{\alpha^2}{a^2}+\frac{\beta^2}{b^2}.$$

(ii) Two tangents are drawn from a point $T(h,k)$ intersects $\mathcal{E}$ at A and B. Show that the equation of AB is

$$\frac{hx}{a^2}+\frac{ky}{b^2}=1.$$

(b) If T lies on the circle $x^2+y^2=a^2+b^2$, show that P lies on the curve

$$x^2+y^2=(a^2+b^2)\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}\right)^2.$$

Ans:

(a) (i) Let $A=(x_1,y_1)$ and $B=(x_2,y_2)$. Then, $x_1+x_2=2\alpha$ and $y_1+y_2=2\beta$. Furthermore, $\dfrac{x_i^2}{a^2}+\dfrac{y_i^2}{b^2}=1$, for $i=1,2$. Therefore,

$$\begin{aligned} \frac{1}{a^2}(x_1^2-x_2^2)+\frac{1}{b^2}(y_1^2-y_2^2) &= 0 \\ \frac{1}{a^2}(x_1+x_2)(x_1-x_2)+\frac{1}{b^2}(y_1+y_2)(y_1-y_2) &= 0 \\ \frac{2\alpha}{a^2}(x_1-x_2)+\frac{2\beta}{b^2}(y_1-y_2) &= 0 \\ \frac{y_1-y_2}{x_1-x_2} &= -\frac{b^2\alpha}{a^2\beta} \end{aligned}$$

The required equation is

$$\begin{aligned} \frac{y-\beta}{x-\alpha} &= \frac{y_1-y_2}{x_1-x_2} \\ \frac{y-\beta}{x-\alpha} &= -\frac{b^2\alpha}{a^2\beta} \\ b^2\alpha x+a^2\beta y &= a^2\beta^2+b^2\alpha^2 \\ \frac{\alpha x}{a^2}+\frac{\beta y}{b^2} &= \frac{\alpha^2}{a^2}+\frac{\beta^2}{b^2} \end{aligned}$$

(ii) Differentiate both sides of the equation of $\mathcal{E}$ with respect to x.

$$\begin{aligned} \frac{2x}{a^2} + \frac{2y}{b^2}\frac{dy}{dx} &= 0 \\ \frac{dy}{dx} &= -\frac{b^2x}{a^2y} \end{aligned}$$

Therefore, the equation of TA and TB are $b^2x_1x + a^2y_1y = a^2b^2$ and $b^2x_2x + a^2y_2y = a^2b^2$, respectively. Since T lies on TA and TB, we have $b^2hx_1 + a^2ky_1 = a^2b^2$ and $b^2hx_2 + a^2ky_2 = a^2b^2$. Therefore, A and B lie on the straight line $b^2hx + a^2ky = a^2b^2$, equivalently $\dfrac{hx}{a^2} + \dfrac{ky}{b^2} = 1$. In other words, it gives the equation of AB.

(b) Rewrite the equation in (a) as

$$\frac{1}{a^2} \cdot \frac{\alpha}{\left(\frac{\alpha^2}{a^2} + \frac{\beta^2}{b^2}\right)}x + \frac{1}{b^2} \cdot \frac{\beta}{\left(\frac{\alpha^2}{a^2} + \frac{\beta^2}{b^2}\right)}y = 1,$$

Since we are considering the chord AB satisfying both conditions in (a)(i) and (a)(ii), the equation of (a)(i) and (a)(ii) are identical. It implies $h = \dfrac{\alpha}{\left(\frac{\alpha^2}{a^2} + \frac{\beta^2}{b^2}\right)}$ and $k = \dfrac{\beta}{\left(\frac{\alpha^2}{a^2} + \frac{\beta^2}{b^2}\right)}$.

Since $T(h,k)$ lies on the circle $x^2 + y^2 = a^2 + b^2$, we have

$$\begin{aligned} h^2 + k^2 &= a^2 + b^2 \\ \left(\frac{\alpha}{\left(\frac{\alpha^2}{a^2} + \frac{\beta^2}{b^2}\right)}\right)^2 + \left(\frac{\beta}{\left(\frac{\alpha^2}{a^2} + \frac{\beta^2}{b^2}\right)}\right)^2 &= a^2 + b^2 \\ \alpha^2 + \beta^2 &= (a^2 + b^2)\left(\frac{\alpha^2}{a^2} + \frac{\beta^2}{b^2}\right)^2 \end{aligned}$$

which gives an equation in α and β. The result follows by replacing α and β with x and y.

Challenge Questions

7. Given an ellipse $\mathcal{E} : \dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$.

(a) Find the equation of tangents to $\mathcal{E}$ which are parallel to the line $y = mx$.
Hence, prove that if the tangent is drawn from a point (h,k), then

$$(a^2 - h^2)m^2 + 2hkm + (b^2 - k^2) = 0.$$

(b) Two non-parallel lines of slopes m_1 and m_2 are equally inclined to the line $y = (\tan\alpha)x$. Show that

$$(1 - \tan^2\alpha)(m_1 + m_2) + 2(m_1m_2 - 1)\tan\alpha = 0.$$

(c) Using the results of (a) and (b), prove that any point from which the tangents drawn to $\mathcal{E}$ are equally inclined to the line $y = (\tan\alpha)x$ lies on

$$x^2 - 2xy\cot 2\alpha - y^2 = a^2 - b^2.$$

Ans:

(a) Let the required tangent line be $y = mx + c$. Substitute it into the equation of $\mathcal{E}$.

$$\begin{aligned} \frac{x^2}{a^2} + \frac{(mx+c)^2}{b^2} &= 1 \\ b^2x^2 + a^2(mx+c)^2 &= a^2b^2 \\ (a^2m^2 + b^2)x^2 + 2a^2mcx + (a^2c^2 - a^2b^2) &= 0 \end{aligned}$$

Since the tangent line only intersects $\mathcal{E}$ at a single point, we have

$$\begin{aligned}
\Delta &= 0 \\
(2a^2mc)^2 - 4(a^2m^2+b^2)(a^2c^2-a^2b^2) &= 0 \\
4a^2b^2c^2 &= 4a^2b^2(a^2m^2+b^2) \\
c &= \pm\sqrt{a^2m^2+b^2}
\end{aligned}$$

Therefore, $y = mx + \pm\sqrt{a^2m^2+b^2}$ are the required lines.

Put $x = h$ and $y = k$ into $y = mx \pm \sqrt{a^2m^2+b^2}$,

$$\begin{aligned}
k &= mh \pm \sqrt{a^2m^2+b^2} \\
k - mh &= \pm\sqrt{a^2m^2+b^2} \\
(k-mh)^2 &= a^2m^2+b^2 \\
(a^2-h^2)m^2 + 2hkm + (b^2-k^2) &= 0
\end{aligned}$$

(b) We have

$$\begin{aligned}
\frac{m_1 - \tan\alpha}{1+m_1\tan\alpha} &= \frac{\tan\alpha - m_2}{1+m_2\tan\alpha} \\
(1-\tan^2\alpha)(m_1+m_2) + 2(m_1m_2-1)\tan\alpha &= 0
\end{aligned}$$

(c) From (a),

$$m_1 + m_2 = -\frac{2hk}{a^2-h^2} \qquad \text{and} \qquad m_1m_2 = \frac{b^2-k^2}{a^2-h^2}.$$

Substitute them into the equation obtained in (b), we have

$$\begin{aligned}
(1-\tan^2\alpha)\left(-\frac{2hk}{a^2-h^2}\right) + 2\left(\frac{b^2-k^2}{a^2-h^2} - 1\right)\tan\alpha &= 0 \\
\cot 2\alpha\left(-\frac{2hk}{a^2-h^2}\right) + \left(\frac{b^2-k^2}{a^2-h^2} - 1\right) &= 0 \\
-2hk\cot 2\alpha + (b^2-k^2) + (a^2-h^2) &= 0 \\
h^2 + 2hk\cot 2\alpha + (b^2-k^2) + k^2 &= a^2+b^2
\end{aligned}$$

which gives an equation in h and k. The result follows by replacing h and k with x and y.

8. Let $O = (0,0)$ and let F_1 and F_2 be the foci of the hyperbola $\mathcal{H} : \dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$. Suppose that P is a point lying on $\mathcal{H}$ and the tangent of $\mathcal{H}$ at P intersects the asymptotes of $\mathcal{H}$ at A and B.

(a) Show that
   (i) AB is bisected at the point P;
   (ii) the area of triangle OAB is ab.

(b) Prove that

$$OA \cdot OB = a^2 + b^2.$$

Hence, show that A, B, F_1 and F_2 are concyclic.

Ans:

(a) (i) Let $P = (a\sec\theta, b\tan\theta)$ be a point on $\mathcal{H}$ The equation of tangent at P is

$$\frac{x}{a}\sec\theta - \frac{y}{b}\tan\theta = 1.$$

The intersection point of the tangent line and the asymptote $y = \frac{b}{a}x$ is $A = \left(\frac{a\cos\theta}{1-\sin\theta}, \frac{b\cos\theta}{1-\sin\theta}\right)$ and the intersection point of the tangent line and the asymptote $y = -\frac{b}{a}x$ is $B = \left(\frac{a\cos\theta}{1+\sin\theta}, -\frac{b\cos\theta}{1+\sin\theta}\right)$. The mid-point of AB is

$$\begin{aligned}\left(\frac{a\cos\theta}{2}\left(\frac{1}{1-\sin\theta}+\frac{1}{1+\sin\theta}\right), \frac{b\cos\theta}{2}\left(\frac{1}{1-\sin\theta}-\frac{1}{1+\sin\theta}\right)\right) &= \left(\frac{a\cos\theta}{1-\sin^2\theta}, \frac{b\sin\theta\cos\theta}{1-\sin^2\theta}\right)\\ &= (a\sec\theta, b\tan\theta)\\ &= P\end{aligned}$$

(ii) The area of the triangle OAB is

$$\begin{aligned}\left|\frac{1}{2}\overrightarrow{OA}\times\overrightarrow{OB}\right| &= \left|\frac{1}{2}\begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ \frac{a\cos\theta}{1-\sin\theta} & \frac{b\cos\theta}{1-\sin\theta} & 0\\ \frac{a\cos\theta}{1+\sin\theta} & -\frac{b\cos\theta}{1+\sin\theta} & 0\end{vmatrix}\right|\\ &= \frac{1}{2}\left|\frac{-2ab\cos^2\theta}{1-\sin^2\theta}\right|\\ &= ab\end{aligned}$$

(b)

$$\begin{aligned}OA^2\cdot OB^2 &= \left[\left(\frac{a\cos\theta}{1-\sin\theta}\right)^2+\left(\frac{b\cos\theta}{1-\sin\theta}\right)^2\right]\cdot\left[\left(\frac{a\cos\theta}{1+\sin\theta}\right)^2+\left(-\frac{b\cos\theta}{1+\sin\theta}\right)^2\right]\\ &= \frac{(a^2+b^2)\cos^2\theta}{(1-\sin\theta)^2}\cdot\frac{(a^2+b^2)\cos^2\theta}{(1+\sin\theta)^2}\\ &= (a^2+b^2)^2\\ OA\cdot OB &= a^2+b^2\end{aligned}$$

Let X be the intersection of AB and F_1F_2. Since X lies on the x-axis, we have $X = (a\cos\theta, 0)$. Now

$$\begin{aligned}\overrightarrow{XA} &= \left(\frac{a\cos\theta}{1-\sin\theta}-a\cos\theta, \frac{b\cos\theta}{1-\sin\theta}\right) = \left(\frac{a\cos\theta\sin\theta}{1-\sin\theta}, \frac{b\cos\theta}{1-\sin\theta}\right)\\ \overrightarrow{XB} &= \left(\frac{a\cos\theta}{1+\sin\theta}-a\cos\theta, \frac{-b\cos\theta}{1+\sin\theta}\right) = \left(\frac{-a\cos\theta\sin\theta}{1+\sin\theta}, \frac{-b\cos\theta}{1+\sin\theta}\right).\end{aligned}$$

Since $\overrightarrow{XA}$, $\overrightarrow{XB}$ point in opposite direction, we have

$$\begin{aligned}XA\cdot XB &= -\overrightarrow{XA}\cdot\overrightarrow{XB}\\ &= -\left(\frac{-a^2\cos^2\theta\sin^2\theta}{1-\sin^2\theta}-\frac{b^2\cos^2\theta}{1-\sin^2\theta}\right)\\ &= a^2\sin^2\theta+b^2.\end{aligned}$$

On the other hand,

$$\begin{aligned}XF_1\cdot XF_2 &= (ae-a\cos\theta)(ae+a\cos\theta) = a^2e^2-a^2\cos^2\theta\\ &= a^2+b^2-a^2\cos^2\theta = a^2\sin^2\theta+b^2\\ &= XA\cdot XB.\end{aligned}$$

By the intersecting chords theorem, A, B, F_1 and F_2 are concyclic.

Alternative method: Let B' be the reflection of B about the y-axis. Since OA and OB make the same angle with the x-axis, A, O, B' lies on a straight line. Now the line segments AB' and F_1F_2 intersect at O and satisfy that

$$OF_1 \cdot OF_2 = (ae - 0)(0 - (-ae)) = a^2e^2 = a^2 + b^2 = OA \cdot OB = OA \cdot OB'.$$

By the intersecting chords theorem, A, B', F_1 and F_2 are concyclic. Since the centre of the circle lies on the y-axis, B lies on the same circle by symmetry.

Optional Questions

9. Let $\mathcal{E} : \dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ be an ellipse.
Suppose that F_1 and F_2 are foci of the ellipse and P is a point on $\mathcal{E}$. Show that

$$PF_1 + PF_2 = 2a.$$

(Hint: Recall the definition of eccentricity.)

Ans:

Let E_1 and E_2 be the perpendicular feet of P on the directrices $x = -\dfrac{a}{e}$ and $x = \dfrac{a}{e}$, respectively and let e be the eccentricity of $\mathcal{E}$. Then,

$$PF_1 + PF_2 = ePE_1 + ePE_2 = eE_1E_2 = e\left(\frac{2a}{e}\right) = 2a.$$

10. Let $\mathcal{H} : xy = 1$, where $x, y > 0$, be a part of a hyperbola. Let $S = \left(s, \dfrac{1}{s}\right)$ and $T = \left(t, \dfrac{1}{t}\right)$ be distinct points on $\mathcal{H}$.

(a) Show that the equation of the chord ST is $x + sty = s + t$.

(b) Suppose that ST passes through $(2, 2)$.

(i) Show that $t = \dfrac{s-2}{2s-1}$ and find all possible values of s.

(ii) If M is the mid-point of ST, show that the equation of locus of M is $(x-1)(y-1) = 1$.

Ans:

(a) The equation of the chord ST is

$$\begin{aligned} \frac{y - \frac{1}{s}}{x - s} &= \frac{\frac{1}{s} - \frac{1}{t}}{s - t} \\ \frac{y - \frac{1}{s}}{x - s} &= -\frac{1}{st} \\ x + sty &= s + t \end{aligned}$$

(b) (i) Put $x = 2$ and $y = 2$ into the equation obtained in (a),

$$\begin{aligned} 2 + 2st &= s + t \\ t &= \frac{s-2}{2s-1} \end{aligned}$$

Since $\dfrac{s-2}{2s-1} = t > 0$ and $s > 0$, we have $0 < s < \dfrac{1}{2}$ or $s > 2$.

(ii) $M = \left(\frac{s+t}{2}, \frac{\frac{1}{s}+\frac{1}{t}}{2}\right) = \left(\frac{s+t}{2}, \frac{s+t}{2st}\right)$. Let $x = \frac{s+t}{2}$ and $y = \frac{s+t}{2st}$ From (b)(i), we have $2 + 2st = s + t$, so $x = 1 + st$ and $y = \frac{1+st}{st}$.

Put $st = x - 1$ into $y = \frac{1+st}{st}$,

$$\begin{aligned} y &= \frac{1+(x-1)}{(x-1)} \\ xy - y &= x \\ xy - y - x + 1 &= 1 \\ (x-1)(y-1) &= 1 \end{aligned}$$

11. Let $P = (p, q)$ be a point which does not lie on the parabola $\mathcal{P} : y^2 = 4x$.

(a) If $y = mx + c$ is a straight line passing through the point P tangent to the parabola $\mathcal{P}$, show that $q = pm + \frac{1}{m}$.

(b) Given that the two tangents for the parabola $\mathcal{P}$ with slopes m_1 and m_2 intersects at the point P.

(i) Prove that $\left(\frac{1}{m_1} - \frac{1}{m_2}\right)^2 = q^2 - 4p$.

(ii) If $\left(\frac{1}{m_1} - \frac{1}{m_2}\right)^2 = 5$ and the point P lies on the line $y = x$, find the coordinates of P.

(c) If the tangents PT_1 and PT_2 are drawn to the parabola $\mathcal{P}$ where T_1 and T_2 are the points of contact. Let $\angle T_2PT_2 = \alpha$, where $\alpha \neq \frac{\pi}{2}$. Prove that the equation of the locus of the point P is

$$y^2 - 4x = (x+1)^2 \tan^2 \alpha.$$

What happens if $\alpha = \frac{\pi}{2}$?

Ans:

(a) Put $y = mx + c$ into the equation of $\mathcal{P}$,

$$\begin{aligned} (mx+c)^2 &= 4x \\ m^2x^2 + (2mc-4)x + c^2 &= 0 \end{aligned}$$

Since the tangent line only intersects $\mathcal{P}$ at a single point, we have

$$\begin{aligned} \Delta &= 0 \\ (2mc-4)^2 - 4(m^2)(c^2) &= 0 \\ -16mc + 16 &= 0 \\ c &= \frac{1}{m} \end{aligned}$$

Therefore, the tangent is $y = mx + \frac{1}{m}$. Furthermore, it passes through the point $P = (p, q)$. Therefore, $q = pm + \frac{1}{m}$.

(b) (i) Rewriting the equation obtained in (a), we have $pm^2 - qm + 1 = 0$. Then, m_1 and m_2 are

the roots of this equation, so $m_1 + m_2 = \dfrac{q}{p}$ and $m_1 m_2 = \dfrac{1}{p}$. Therefore,

$$\begin{aligned}
\left(\frac{1}{m_1} - \frac{1}{m_2}\right)^2 &= \frac{(m_1 - m_2)^2}{(m_1 m_2)^2} \\
&= \frac{(m_1 + m_2)^2 - 4m_1 m_2}{(m_1 m_2)^2} \\
&= \frac{(\frac{q}{p})^2 - 4(\frac{1}{p})}{(\frac{1}{p})^2} \\
&= q^2 - 4p
\end{aligned}$$

(ii) We have $q^2 - 4p = \left(\dfrac{1}{m_1} - \dfrac{1}{m_2}\right)^2 = 5$ and $p = q$.

It implies $p = q = -1$ or $p = q = 5$, i.e. $P = (-1, -1)$ or $P = (5, 5)$.

(c)

$$\begin{aligned}
|\tan\alpha| &= \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right| \\
\tan^2\alpha &= \frac{(m_1 - m_2)^2}{(1 + m_1 m_2)^2} \\
&= \frac{(m_1 + m_2)^2 - 4m_1 m_2}{(1 + m_1 m_2)^2} \\
&= \frac{(\frac{q}{p})^2 - 4(\frac{1}{p})}{(1 + \frac{1}{p})^2} \\
&= \frac{q^2 - 4p}{(p+1)^2} \\
q^2 - 4p &= (p+1)^2 \tan^2\alpha
\end{aligned}$$

which gives an equation in p and q. The result follows by replacing p and q with x and y.

When $\alpha = \dfrac{\pi}{2}$, we have $\dfrac{1}{p} = m_1 m_2 = -1$, i.e. $p = -1$. Therefore, P moves along the vertical line $x = -1$, which is exactly the directrix of $\mathcal{P}$.